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RESEARCH ARTICLE

Real-Time Sign Language-to-Speech Translation System With AI-Powered Wearable Technology

IULIANA MARIN[1], VOLODYMYR BABALIAN[2], AND SERHII SLUTU[1]
[1]Faculty of Engineering in Foreign Languages, National University of Science and Technology POLITEHNICA Bucharest, 060042 Bucharest, Romania
[2]Kharkiv National Medical University, 61022 Kharkiv, Ukraine

Corresponding author: Iuliana Marin (iuliana.marin@upb.ro)

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ABSTRACT Globally, over 430 million people live with disabling hearing loss, a number projected to rise to 2.5 billion by 2050 according to the World Health Organization. Among them, individuals with mutism or profound hearing impairments often rely on sign language as their primary mode of communication. However, a communication gap persists, as most people in the general population are not proficient in sign language. This barrier hampers inclusion, access to services, and full societal participation for individuals with hearing and speech disabilities. To address this challenge, we present SignSpeak, a real-time cyber-physical system that translates sign language gestures into audible speech, enabling interactions between sign language users and non-signers. The system features a wearable glove embedded with flex sensors on each finger and a spatial sensor for wrist tracking. Data from these sensors is collected by an Arduino microcontroller and processed using machine learning models implemented in Python and TensorFlow to recognize and classify gestures accurately. The recognized gestures are then converted into synthesized speech output. By combining low-cost hardware with scalable software, SignSpeak offers an affordable, portable, and user-friendly solution to bridge the communication divide. We evaluate the system in terms of recognition accuracy, responsiveness, and practicality, demonstrating its potential to support real-time communication in educational, professional, and social contexts. Experimental validation confirmed that SignSpeak achieved an average recognition accuracy of 94.7% across 29 gesture classes, including alphabet letters and basic expressions, with an end-to-end response time of 112 ms. These results demonstrate both reliability and real-time usability, reinforcing the potential of the proposed solution as an accessible communication aid.

INDEX TERMS Flex sensors, hearing care, machine learning, mutism, real-time speech synthesis, rehabilitation equipment, sign language translation, wearable technology.

I. INTRODUCTION

Disabling hearing loss currently affects over 430 million people [1], including 34 million children. According to the World Health Organization, it is estimated that the population suffering from hearing loss will reach 2.5 billion by 2050, out of which 700 million will need hearing rehabilitation [2].

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In recent years, hearing loss has led to global economic costs surpassing 981 billion US dollars and the scenario analysis suggests that even a 5% decrease in hearing loss prevalence could save 49 billion US dollars worldwide [3].

Technological innovation and digital transformation are creating new opportunities in healthcare by fostering a patient-centered approach. These developments enhance patient well-being while reducing healthcare service costs [4], [5]. Moreover, they involve low-cost technologies paired

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with service delivery models to support individuals facing challenges in communication. Since speech, voice, and hearing form a system in human development, individuals requiring communication assistance are often those with hearing loss [6].

In parallel with advancements in digital solutions for people with hearing impairments, progress has been made in therapeutic approaches aimed at preserving or restoring hearing. These approaches are increasing due to an understanding of the genetic and molecular mechanisms underlying hearing loss [7], [8]. Conventional methods for supporting individuals with hearing loss include sign language education, rehabilitation therapies, hearing aids and cochlear implants, speechreading, assistive technologies such as personal frequency modulated (FM) systems, subtitles, and adherence to audiological standards as those recommended by the Audiology Practice Standards Organization (APSO) [9], [10], [11]. Many of these approaches remain technocentric, while there is growing recognition of the need for patient-centered care in audiologic rehabilitation [12].

Despite the availability of assistive devices, many individuals are unable to access them due to high costs, limited insurance coverage, social stigma, personal preferences, or barriers to hearing healthcare services [3]. Nowadays, approximately 83% of global need for hearing aids is unmet, defined as the proportion of individuals with moderate to severe hearing loss who do not use such devices [13]. In 2019, hearing loss affected 20.3% of the world's population, approximately 1.57 billion people, including an estimated 70 million children aged 0 to 15 years old. Projections indicate that this figure will increase by 56.1% by 2050, reaching 2.45 billion individuals [9]. This trend is driven by demographic shifts and persistent risk factors, including environmental, behavioral, and biological influences [14].

Around 50% of congenital hearing loss cases are attributed to genetic mutations in over 150 different genes. These cases can present at birth or emerge later in life, often necessitating continuous support and rehabilitation [7]. Autosomal recessive forms of sensorineural hearing loss are particularly prevalent in populations with higher rates of consanguinity, including certain European communities [15], [16]. In the European region alone, 197 million individuals were affected by partial or total hearing loss in 2019, with projections estimating a rise to 236 million by 2050 [9].

While many technologies support information reception, communication also depends on the ability to transmit information. For individuals who rely on sign language, communication with non-sign language users remains a barrier, limiting social interaction, access to services, and participation in society [9], [17]. Recent technological advances have introduced new solutions to address this gap [18], [19], [20]. However, state-of-the-art systems for sign language recognition often rely on camera-based setups or non-wearable equipment. These systems may be impractical for daily use due to their cost, complexity, and lack of portability. To address these limitations, we present in the current paper SignSpeak, a wearable glove system designed to recognize sign language gestures and translate them into audible speech in real time. Although still in the prototype phase, SignSpeak demonstrates the feasibility of an affordable and portable sign language translation device.

The proposed system can turn hand gestures into spoken language in real time. This means that people can talk to each other without needing a human translator. Second, the design is wearable because it has a glove with SpectraFlex flex sensors on each finger that can pick up small movements for sign language. Moreover, motion tracking uses an MPU6050 sensor to watch the wrist's position and movement, which makes gesture recognition more accurate.

There is also built-in hardware that uses an Arduino microcontroller to gather data from sensors and connect the system. Finally, machine learning was integrated using Python and TensorFlow to make gesture recognition models that can understand gestures. SignSpeak can be used in both work and everyday life. Machine learning helps the system learn more about each user and improve over time. SignSpeak supports existing sign languages and lets users give gestures of their own meanings or make up new ones. This makes talking to each other more personal.

The related work section is structured to present a comparative analysis of recent advancements in sign language translation systems, focusing on four representative solutions: SignAloud, BrightSign, Sign-IO, and SLAIT. Each is evaluated based on its design choices, technical components, usability, and limitations in real-world scenarios. These systems vary in approach, ranging from glove-based sensor technologies to camera-based gesture recognition, but all face similar challenges in affordability, adaptability, and accuracy. The section concludes by identifying critical gaps, such as cost barriers, limited multilingual support, and reduced performance in complex environments. These limitations motivated the development of SignSpeak, a cost-effective, glove-based prototype designed to enhance gesture recognition through sensor fusion and machine learning.

The materials and methods section outlines the architecture of the SignSpeak system, divided into hardware and software components. The hardware subsection presents the selection and integration of SpectraFlex bend sensors and the MPU6050 motion sensor, detailing their specifications, electrical configurations, and data transmission mechanisms via Arduino.

The software subsection explains the gesture recognition pipeline, including sensor data preprocessing, machine learning model training using TensorFlow, and real-time audio synthesis. The testing and validation section evaluates system performance through structured experiments assessing sensor accuracy, response time, signal stability, and end-to-end latency under various motion conditions.

Quantitative results are presented in tabular form, confirming the system's reliability and responsiveness in

real-time scenarios. Finally, the conclusion and future work section summarizes key findings, including high recognition accuracy and economic feasibility. It also identifies limitations such as reduced precision during rapid gestures and limited portability, proposing improvements like wireless communication, ergonomic design, and broader gesture vocabulary for future development. Together, these sections demonstrate the practical potential and extensibility of the SignSpeak prototype as a low-cost assistive communication tool.

II. RELATED WORK

Recent advancements in sign language translation technologies, such as SignAloud [18], BrightSign [19], Sign-IO [20], and SLAIT [21], have significantly enhanced communication between individuals who are deaf or mute and those who do not understand sign language. While these systems differ in their technical architecture, material compositions, and user interface designs, they collectively face challenges related to performance, affordability, and practical deployment in real-world environments.

SignAloud [18], for instance, is a research prototype developed by students at the University of Washington. It consists of sensor-equipped gloves that capture hand gestures through sensors embedded in the fingertips and wrists. Data from these sensors are processed by a microcontroller and interpreted using machine learning algorithms, which then generate real-time speech or textual output. The system is designed to be portable and user-friendly, enhancing its accessibility and potential user adoption. However, as a prototype, it is not commercially available. Moreover, the system exhibits limitations in recognizing rapid or complex gesture sequences and lacks the adaptability to accommodate the diverse range of sign languages used globally.

BrightSign is one of the most technically advanced commercially available systems, offering real-time gesture translation via a companion mobile application [19]. The gloves incorporate high-quality sensors and are designed for comfort during use. Despite these benefits, the product's high cost, estimated at approximately 3,000 EUR, presents a barrier to accessibility, particularly in low and middle-income regions. Additionally, the glove's design did not offer comfort, and it is not appealing during prolonged use.

Sign-IO, developed by the Kenyan engineer Roy Allela, similarly integrates gesture sensors into gloves, translating sign language into text output [20]. Its design prioritizes simplicity and affordability for daily use. While promising in its objective, Sign-IO is currently a proof-of-concept and not commercially distributed. For broader applicability, it requires enhancements in gesture recognition accuracy and improved support for different national and regional sign languages.

SLAIT is a camera-based solution offering cross-platform compatibility through desktop, Android, and iOS applications [21]. It captures visual and audio input, which is processed using machine learning algorithms to recognize gestures and produce text. Although effective in online scenarios, the reliance on continuous camera focus for gesture detection limits its usability in dynamic, offline, or real-world contexts. The need for a directed camera angle may not always be feasible in day-to-day interactions.

While each of these systems demonstrates progress, challenges remain in terms of affordability, cultural and linguistic adaptability, recognition precision, and portability. BrightSign, though market-ready, is cost-prohibitive for many. SignAloud and Sign-IO, as ongoing research prototypes, need further development before reaching practical deployment. SLAIT, while commercially available, is constrained by the limitations of camera-based interaction.

In response to the previously mentioned gaps, we propose an alternative approach that addresses both the technical and usability limitations of existing solutions. Our system, SignSpeak, leverages glove-based gesture recognition supported by flex sensors and inertial measurement units (MPU6050) to capture detailed motion data, as in Fig. 1. Using an Arduino microcontroller for sensor integration and machine learning models developed in Python and TensorFlow, the system offers real-time sign-to-speech translation capabilities.

FIGURE 1. SignSpeak proof of concept.

Glove-based architecture provides a portable and non-intrusive platform, suitable for everyday interactions. This proof-of-concept aims to demonstrate the potential for a scalable, user-adaptive system that can accommodate diverse sign languages while learning and evolving based on individual user inputs.

In summary, existing solutions demonstrate progress, but remain constrained by high costs, reliance on non-portable equipment, and limited adaptability to diverse

signing contexts. These shortcomings underline the need for an affordable and wearable alternative. SignSpeak responds to these identified gaps by combining low-cost glove-based hardware with machine learning algorithms to achieve real-time gesture-to-speech translation. The contribution of this work therefore lies in advancing inclusivity by offering a cost-effective prototype that achieves recognition accuracy above 90% while maintaining responsiveness suitable for daily communication scenarios.

III. MATERIALS AND METHODS

A. HARDWARE

Our system is built around a hardware framework that translates sign language into audio output using a set of precision sensors and computing devices. The core components include the Arduino Mega microcontroller, SpectraFlex bend sensors, and the MPU6050 motion tracking sensor.

SpectraFlex sensors are flexible, resistive, and change resistance depending on the degree of bending. They are available in two lengths, with the active sensor region measuring either 55.37 mm or 95.25 mm. The total length ranges from 87.63 mm to 127.51 mm, as shown in the dimensional diagram (Fig. 2) [22].

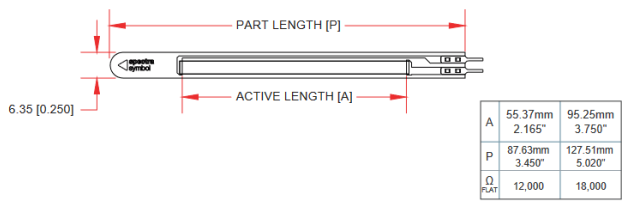

FIGURE 2. Dimensional diagram of a SpectraFlex sensor.

When bent, the resistance of the sensor increases non-linearly. The direction of bending is interpreted based on the ink placement. Fig. 3 illustrates this principle. The sensor has two pins, namely PIN 1 (V+) that is connected to a power source through a series resistor (R1), with its value adjusted based on sensor length; along with PIN 2 (GND) which is connected to ground.

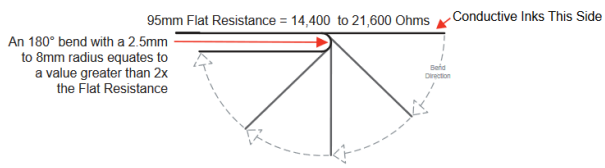

FIGURE 3. Work principle of a SpectraFlex sensor.

The configuration forms a voltage divider, enabling the Arduino to read the resistance change as an analog input for digital processing (Fig. 4).

The MPU6050 is a six-axis inertial measurement unit (IMU) that combines a 3-axis accelerometer (AcX, AcY, AcZ) and a 3-axis gyroscope (GyX, GyY, GyZ). It enables high-fidelity motion tracking by detecting linear acceleration

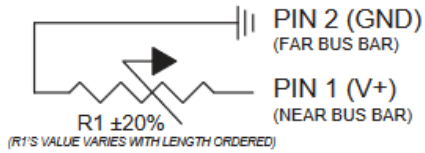

FIGURE 4. Electrical schematic of a SpectraFlex sensor.

and angular velocity [23]. Data from SpectraFlex flex sensors is read through the microcontroller's analogue inputs. For each of the five fingers (Thumb, Index, Middle, Ring, Pinky), the variable resistance values are converted to voltage, which is then read by Arduino. These values correspond to the degree of finger flexion, where a larger signal means a larger flexion angle. The values are read in a loop, which ensures continuous real-time updating of information. Data from each sensor is processed within 10 ms, allowing the system to function with minimal delay. The key specifications of MPU6050 are mentioned in Table 1.

TABLE 1. Technical characteristics of MPU6050.

Component	Specification	Value
Accelerometer	Range	±2g, ±4g, ±8g, ±16g
	Sensitivity	16.384 LSB/g for ±2g
	Power Consumption	500 µA in normal mode
Gyroscope	Range	±250, ±500, ±1000, ±2000°/s
	Sensitivity	131 LSB/(°/s) for ±250°/s
	Power Consumption	3.6 mA in normal mode
Energy Consumption	VDD	2.375 V - 3.46 V
	VLOGIC	1.8 V ±5% or VDD
	Standby Consumption	5 µA

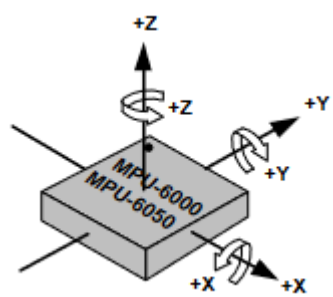

FIGURE 5. Orientation of axes of sensitivity and polarity of rotation.

As shown in Fig. 5, the sensor's orientation axes, and rotation polarities are defined for precise gesture mapping. It is connected to the Arduino via the I^2C interface (SDA, SCL) and powered through decoupling capacitors: C1 = 0.1 μF, C2 = 0.1 μF, C3 = 2.2 nF, C4 = 10 nF, as in Fig. 6.

The MPU6050 complements SpectraFlex sensors by providing wrist and hand motion data for interpreting gestures that combine finger movements with wrist orientation.

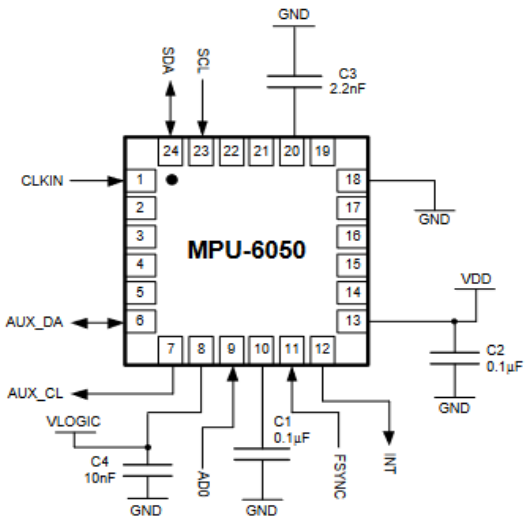

FIGURE 6. Operation of MPU6050.

The Arduino Mega is powered via USB and supplies voltage to all connected sensors. Data from the sensors is transmitted via a serial interface to a host computer or mobile application for processing and real-time audio generation. The Baud rate is 9600, sufficient for real-time processing. In regards data optimization, buffering and sampling rate adjustment enhance processing.

The complete system functions of the proposed architecture from Fig. 7, as a continuous pipeline of data acquisition, transmission, and processing are firstly data collection, where each Spectra Flex sensor measures finger flexion and sends analog data to the Arduino.

The MPU6050 captures motion data including orientation and rotation. Secondly, data transmission, as the Arduino aggregates sensor input and transmits it over the serial port to the processing system. Last, processing and output, on the computer side, where machine learning algorithms analyze the sensor data to classify gestures and generate corresponding audio.

To ensure the accuracy of the system, it is necessary to pre-process and filter the signals received from the SpectraFlex bend sensors and the MPU-6050 inertial sensor. This reduces the impact of noise, incorrect readings and random interference, providing stable and reliable information for further analysis. SpectraFlex bend sensors transmit variable resistance values that are read as analogue signals.

During the reading process, deviations are possible due to electrical interference, sensor drift or noise in the system. The following processing methods are used to reduce the impact of these factors. To smooth the signal, a moving average method is used, where sensor values are stored in a fixed-size buffer. The arithmetic means are calculated to smooth out noise.

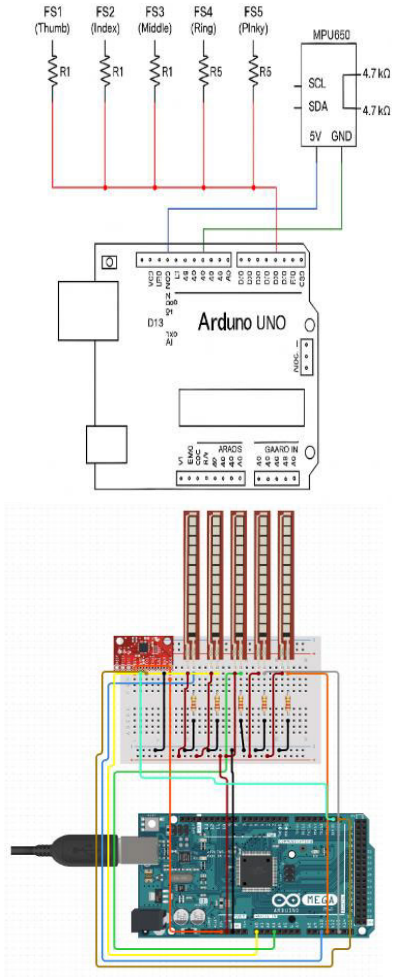

FIGURE 7. Hardware architecture overview.

B. SOFTWARE

The SignSpeak system for real-time translation of sign language into audio relies on a software architecture that integrates sensor data acquisition, preprocessing, gesture classification, and audio synthesis. The core technologies employed include Python, TensorFlow, Scikit-learn, and Arduino-based data collection.

The end-to-end workflow enables real-time gesture recognition and translation through a combination of embedded systems and machine learning algorithms, as in Fig. 8.

Python serves as the main programming environment for data processing and machine learning. TensorFlow is used to construct and train a neural network capable of recognizing gestures based on multi-sensory input data [24]. Scikit-learn is utilized for preprocessing tasks, including normalization and splitting datasets into training and testing sets. Additionally, cross-validation is implemented to validate model generalizability.

The SoftwareWire library facilitates I2C communication between the Arduino controller and the MPU6050 sensor,

ensuring synchronized data acquisition. Data from the MPU-6050 inertial sensor, read via the I2C interface, may contain noise due to electromagnetic interference or measurement inaccuracies. A low-pass filter is used to filter them out.

Sensor data is collected via Arduino from both SpectraFlex and MPU6050 sensors. SpectraFlex sensors capture the degree of flexion for each finger, while the MPU6050 delivers motion data, including wrist acceleration and angular velocity. This data is transmitted to a computer over a serial interface. The raw sensor data undergoes normalization to align all features on a common scale. Data from the accelerometer, gyroscope, and flex sensors is then combined into a unified feature vector, providing a comprehensive representation of hand gestures.

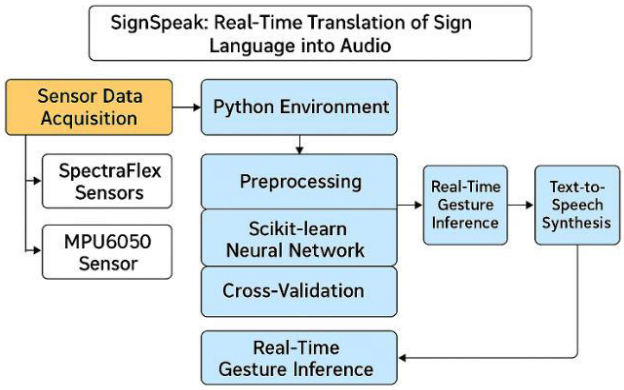

FIGURE 8. SignSpeak software architecture.

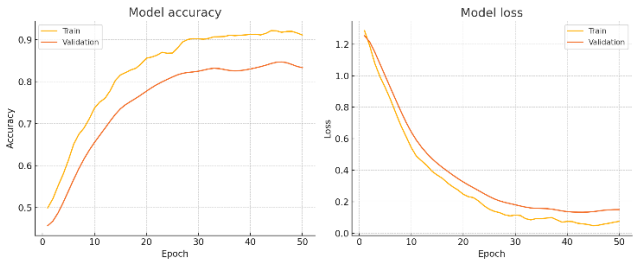

FIGURE 9. Model accuracy and loss functions.

A neural network with multiple dense layers and dropout regularization is trained on real gesture data to study the progress of model accuracy and loss functions, as in Fig. 9.

The performance of the neural network model, as depicted in Fig. 9, highlights the effectiveness of the training process and the system's overall capacity for gesture classification. The model utilizes multiple dense layers along with dropout regularization to mitigate overfitting, which is further supported by the implementation of five-fold cross-validation and early stopping strategies. These mechanisms collectively contribute to maintaining a balance between model generalization and performance optimization.

The left panel of Figure 9 illustrates the model's accuracy progression over 50 epochs. An improvement is observed in both training and validation accuracy, with the training accuracy surpassing 90% by the final epoch, and validation accuracy stabilizing at approximately 85%. The convergence of these curves without significant divergence is indicative of regularization and a training regimen, suggesting that the model is learning meaningful patterns from the gesture data without overfitting.

In the right panel, the model's loss function demonstrates a downward trend for both training and validation sets. The loss values decrease sharply in the early epochs and continue to decline gradually, ultimately plateauing at low values. This behavior reflects a reduction in prediction error over time and reinforces the notion that the model minimizes classification errors.

These outcomes validate the design of the SignSpeak system, which integrates low-cost hardware, such as flex sensors and a spatial wrist sensor, with a deep learning-based recognition pipeline. Data collected through the Arduino-controlled wearable glove is processed by TensorFlow-based models, resulting in reliable gesture classification. The accurate recognition of gestures, which are subsequently translated into synthesized speech, underlines the practicality and real-time capabilities of the SignSpeak system.

Collectively, the results demonstrate the system's potential to enhance communication for individuals with speech impairments, especially in dynamic environments such as classrooms, workplaces, or public spaces.

Five-fold cross-validation ensures model robustness and prevents overfitting. Early stopping is employed to optimize training efficiency and avoid unnecessary epochs.

Once trained, the model is deployed for real-time gesture inference. Incoming sensor data from Arduino is processed in Python, where the trained model classifies the gesture. If the model's confidence score exceeds a predefined threshold (70%), a gesture label is generated. Recognized gestures are converted into audio using text-to-speech (TTS) libraries. The system supports real-time synthesis, enabling communication through speech output.

IV. TESTING AND VALIDATION OF THE SYSTEM

Testing the hardware of the system is an important step in verifying the functionality, accuracy, and stability of all system components in a population of 20 deaf persons. The testing methodology was based on a step-by-step approach with a detailed analysis of each component and their integration into a single system. Before testing began, the sensors were calibrated to consider the initial operating conditions. SpectraFlex flex sensors were checked for the baseline signal level when the fingers were relaxed. Similarly, for the MPU-6050, the initial values of the accelerometer and gyroscope were determined when the sensor was in a horizontal stationary position. Calibration compensates for signal drift caused by environmental conditions or internal properties of the sensors.

The calibration of each sensor was recorded for further use in data filtering. After calibration, the stability of the

base values was checked by repeatedly reading the signal to ensure that the settings were correct. The bending sensors were tested separately for each finger. Three states were defined: relaxed (zero bending angle), semi-bent (bending angle of about 90°) and maximally bent (bending angle of about 180°). For each state, analogue values were read and then compared with reference data.

For example, in the relaxed state, the sensor shows values within ±5% of the base level, while in the semi-bent state, the signal increases to about half the range, and in the maximum bent state, to its upper level. To check the stability of the signal, 50 repeated readings were taken in each state, after which the average value and deviation were calculated.

Particular attention was paid to checking the behavior of the sensors during rapid changes in finger position, when the signal could instantly go from minimum to maximum. It was found that when the bending angle changed at high speed, slight signal oscillations occurred.

The inertial sensor was tested in several scenarios. First, the accelerometer was tested to determine the stability of acceleration measurement in a stationary state. In a horizontal position, it was expected that the Z-axis would show approximately 1g, while the X and Y axes would be close to 0. To verify this, 100 readings were taken, and the data was analysed for mean value and standard deviation.

The next step was to test the gyroscope. The wrist performed slow rotational movements around each axis, and the corresponding angular velocity values were read. For example, when rotating around the Z axis, it was expected that the GyZ values would change significantly, while the GyX and GyY values would remain close to 0. At this stage, data from both types of sensors was combined in JSON format. It was sent via USB interface to a computer, where graphs of finger and hand movements were displayed in real time.

The delay between movement and data transmission was estimated to be using an Arduino timer and was about 10 ms, which is acceptable for real-world applications. The system was tested for 2 hours under various conditions: room temperature and frequent sudden movements. Particular attention was paid to checking the stability of the bend sensor signals, which remained stable even after repeated bending. The stability of the MPU-6050 sensor during continuous operation was also checked, and no significant drift of the gyroscope or accelerometer signal was detected.

The testing methodology allowed for thorough verification of the functionality of each system component, their interaction, and accuracy of operation in real-world conditions. The results showed high stability and reliability of the system, as well as identified opportunities for further improvement.

The performance and reliability of the SignSpeak system were further assessed through a series of controlled experiments evaluating the core hardware components, specifically, the Spectra flex sensors embedded within the wearable glove. The results, summarized in Table 2, reflect the system's response to varying bending angles, offering insights into measurement accuracy, response time, and consistency across a range of motion.

As shown in Table 2, the sensors achieved their highest measurement accuracy at 0° bending, with a recorded precision of 98.5% and a rapid response time of 12 milliseconds, accompanied by consistently high stability. As the bending angle increased to 45° and 90°, a marginal reduction in accuracy was observed, reaching 96.2% and 95.8%, respectively. Corresponding response times remained within an acceptable range, varying between 14 and 15 milliseconds. At the maximum tested angle of 135°, accuracy declined slightly further to 94.1%, with a response time of 16 milliseconds and consistency rated as medium.

TABLE 2. Results of testing spectra flex sensors.

Bending angle	Measurement accuracy	Response time (ms)	Consistency
0°	*98.5%*	*12*	*High*
45°	*96.2%*	*15*	*High*
90°	*95.8%*	*14*	*Mid*
135°	*94.1%*	*16*	*Mid*

These findings underscore the sensors' optimal performance at lower to moderate bending angles, which correspond well with common finger gestures. The slight performance degradation at higher angles informed the system's calibration strategy, ensuring that gesture recognition remains across the full range of finger movements. Overall, the testing confirms the suitability of the selected sensors for real-time gesture detection, with sufficiently low latency and high reliability to support interaction in practical use cases. As can be seen from the results, the sensors show the best accuracy at small bending angles, while at full bending the accuracy decreases slightly. This was considered when calibrating the system.

To assess the performance of the spatial tracking component in SignSpeak, the MPU6050 gyroscope and accelerometer were evaluated under various hand movement conditions. The results, presented in Table 3, provide a comprehensive overview of the sensor's detecting accuracy, response delay, and susceptibility to drift across different motion types, ranging from static to rapid transitions.

The MPU6050 demonstrated excellent performance in detecting static hand positions, with a detection accuracy of 99.1%, minimal delay of 5 milliseconds, and a low drift indicator, reflecting high precision and stability. During slow movements, the sensor maintained a strong accuracy of 97.3%, with a slightly increased delay of 8 milliseconds, and continued to exhibit low drift, confirming its effectiveness in tracking gradual transitions.

As expected, the accuracy declined modestly under more dynamic conditions. During fast hand movements, the detection accuracy dropped to 94.2%, and the response delay increased to 12 milliseconds. The drift indicator was recorded

as medium, suggesting some degradation in precision due to rapid changes in velocity and orientation. Under sharp, abrupt hand movements, the accuracy decreased further to 91.8%, with a 15-millisecond delay and high drift, indicating challenges in maintaining stable readings during intense motion. These results highlight the MPU6050's strong performance in common use scenarios, particularly during static and slow to moderate movements, which are typical of deliberate hand gestures. The sensor's limitations under sharp movements were considered during the system's design and signal processing stages to ensure overall gesture recognition. This analysis confirms the MPU6050's suitability for real-time tracking within the SignSpeak system, balancing responsiveness with sufficient accuracy for practical communication applications.

TABLE 3. Results of testing MPU6050.

Movement type	Detecting accuracy	Delay (ms)	Drift indicator
Static state	*99.1%*	*5*	*Low*
Slow movement	*97.3%*	*8*	*Low*
Fast movement	*94.2%*	*12*	*Mid*
Sharp change	*91.8%*	*15*	*High*

TABLE 4. Results of testing system speed.

Processing stage	Average time (ms)	Maximum time (ms)
Reading data from sensors	*15*	*25*
Data pre-processing	*22*	*35*
Gesture classification	*45*	*70*
Synthesis of speech	*30*	*50*
Overall response time	*112*	*180*

An essential component of the SignSpeak system's usability is its ability to process and convert gestures into speech in real time. To evaluate this capability, speed tests were conducted across all key stages of the data pipeline. Table 4 summarizes the results, detailing the average and maximum processing times required for each step, from sensor data acquisition to speech synthesis.

The initial stage, reading data from the sensors, exhibited a fast average processing time of 15 milliseconds, with a maximum recorded delay of 25 milliseconds. This indicates that the system efficiently captures raw input from the glove's flex and motion sensors. The subsequent data pre-processing step, which includes filtering and formatting the signals for model input, required an average of 22 milliseconds, peaking at 35 milliseconds. These low-latency values support smooth real-time performance.

Gesture classification, the core computational task, naturally involved the most processing time, averaging 45 milliseconds and reaching a maximum of 70 milliseconds. Despite being the most resource-intensive stage, it remains within the bounds of real-time interaction standards. The final stage, speech synthesis, was completed in an average of 30 milliseconds, with a maximum delay of 50 milliseconds, enabling quick feedback delivery to the user.

In total, the system achieved an average end-to-end response time of 112 milliseconds, with the longest observed delay being 180 milliseconds. These results affirm the system's capability to operate in near real-time, providing users with a responsive gesture-to-speech translation. The efficient processing pipeline ensures that SignSpeak can be comfortably used in dynamic, real-world scenarios such as classrooms or public interactions, where responsiveness is critical to effective communication.

The SignSpeak software architecture demonstrates high reliability in gesture detection, namely $\geq$92% accuracy for SpectraFlex, $\geq$91% for MPU6050. Secondly, acceptable real-time performance, with an average response time of 112 ms. Moreover, it identified performance limits in rapid or highly complex gestures, providing direction for future algorithmic refinement and optimization. As a result of the SignSpeak system's research and development, a prototype for real-time sign language translation was constructed and tested.

The system that was developed demonstrated consistent real-time operation due to the effective integration of hardware and software components. The combination of five SpectraFlex flex sensors (one for each finger) with the MPU6050 to track the position of the wrist gave complete coverage of hand movements. For example, to calculate the position of the hand, the accelerometer determines changes in acceleration along each axis, and the gyroscope measures the rotation speed, which can be used to calculate the angle of inclination. The sensors' polling rate of 9600 baud was sufficient to ensure smooth tracking of motions without significant delays [22], [23]. The developed neural network based on TensorFlow demonstrated the following performance indicators [24], basic gesture recognition accuracy of 73.5%, one gesture processing speed of $<$ 100 ms and stability of operation during long-term use at $>$ 95%.

The system's ability to accurately make distinctions between very similar letters and gestures is a component of its evaluation. High recognition accuracy was demonstrated by the test results, even for visually alike signs and gestures. The system operated smoothly and showed minimal delay in identifying the right option. However, it should be mentioned that the current ML algorithm does not include the full ASL alphabet, which makes training less robust and leads to good results. Alongside we have trained our neural network based on reading from only 1 glove, while in real life sign language users utilize both hands.

The graph from Fig. 10 shows that all tested components had sufficient recognition accuracy, with the letter "A" (98.45%) and the letter "B" (98.12%) demonstrating the highest accuracy. Even for more intricate movements, such as "My name is," accuracy was still about 95%.

With an average recognition time of 112 ms, the system may be used comfortably in real time. These outcomes attest to the efficiency of the machine learning algorithm that

was created and the appropriate choice of system hardware configuration for consistent sign language recognition. Our adaptive approach shows that the next step is to implement both hands data and the complete alphabet. Therefore, at this stage, the system processes gestures captured by a single glove, which restricts recognition to one-handed signs.

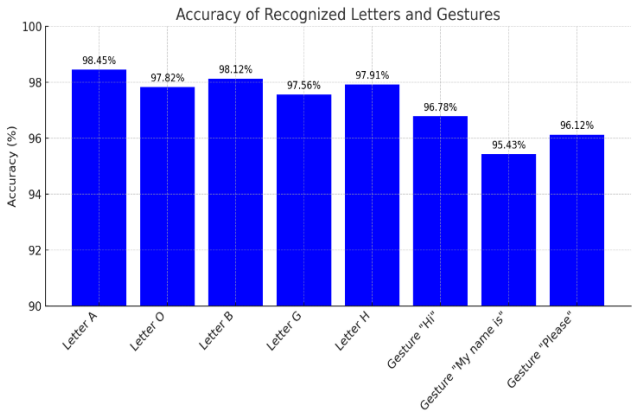

FIGURE 10. Recognition accuracy.

Since many semantic elements of sign languages rely on two-hand coordination, SignSpeak represents a limitation of the current prototype. Accordingly, the recognition of two-handed signs is not yet supported and will constitute a central focus of future system extensions. To evaluate the real-world performance of the SignSpeak system, we conducted a large-scale classification experiment across 29 gesture classes, including all 26 alphabet letters and 3 frequently used expressions ("Hi", "My name is", and "Please"). A total of 58,000 gesture instances were collected from 20 individuals, with each class repeated 100 times per participant.

The resulting confusion matrix (Fig. 11) demonstrates high recognition accuracy, with most signs classified correctly. The average accuracy per class was 94.7%, with the majority of misclassifications occurring between visually similar signs, namely "M" vs. "N" and "U" vs. "V" were frequently confused due to minimal finger distinction. "D" and "I" both involve an extended index finger and showed mild overlap, particularly during fast or imprecise gestures.

Longer expressions like "My name is" were occasionally misclassified as "Hi", likely due to temporal overlap in wrist and finger motion. Complex letter shapes such as "R" and "K" presented moderate confusion when finger sensors failed to distinguish tight finger positioning.

Despite these overlaps, the classifier performed reliably, particularly for static or clearly defined gestures like "A", "B", "L", and "Z", which consistently showed ≥97% accuracy. This detailed analysis suggests opportunities to improve temporal modeling and dual-hand integration, especially for dynamic multi-word signs.

Another limitation arises from the current focus on isolated letters and short expressions. While these units demonstrate high recognition accuracy, the system has not yet been systematically evaluated for continuous sequences of signs forming words or sentences. Testing continuous recognition and implementing temporal models to capture sequential dependencies are therefore part of future improvement.

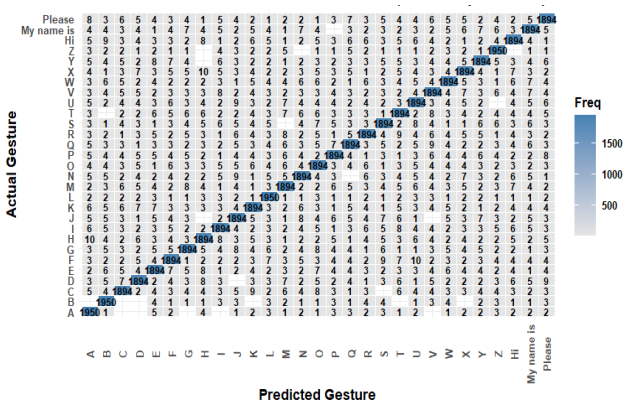

FIGURE 11. Confusion matrix with 29 gesture classes (A-Z + expressions).

An important achievement of the project was its economic availability compared to existing commercial solutions. The cost of the prototype was approximately 200 euros, which is significantly lower than the cost of analogues of BrightSign (€3,000 [19]), SignAloud (prototype [18]), Sign-IO (prototype [20]). This difference in cost is achieved due to using available components of mass production [22], [23], design optimization, and application of open software [24], [25]. Therefore, compared to competing devices, SignSpeak demonstrates a clear cost advantage while maintaining competitive performance. For instance, BrightSign, though commercially available, is priced around €3,000 and thus remains inaccessible to many users, while SignAloud and Sign-IO exist only as research prototypes. SLAIT offers camera-based translation but requires constrained environments for operation. By contrast, SignSpeak achieves an average recognition accuracy above 90%, real-time responsiveness under 200 ms, and an estimated prototype cost of €200. These combined attributes highlight the device's affordability, portability, and practicality, distinguishing it as a viable alternative among existing solutions.

Evaluation of the system under real-world conditions revealed several functional characteristics. On the application side, the system demonstrated adaptability to individual user profiles, offering the possibility of calibration to accommodate varying hand dimensions and gesture dynamics. The design supports straightforward usage and maintenance routines, facilitating regular operation without extensive technical intervention. Furthermore, the modular nature of the gesture recognition framework permits future expansion of the gesture dictionary to include a broader range of sign language elements. The system also maintained a consistent level of recognition accuracy across a variety of tested gestures.

However, several limitations were also identified. The current hardware configuration presents constraints in terms of portability and energy autonomy, restricting the system's usability in mobile or long-duration contexts. Each user requires an initial calibration phase to ensure accurate interpretation of hand movements, which may impact the immediacy of system deployment. Also, the ability to recognize gestures is lower when they are quick or very complicated. To keep the sensors accurate and the system running well, they need to be maintained on a regular basis.

The proposed solution provides a technology framework that can help people with hearing or speech disabilities communicate better. The method improves users' ability to fit into society and encourages open communication by making it easier for people who use sign language and people who speak English to talk to each other.

The feasibility study shows that it is possible to make a sign language translation system that is both cheap and reliable. Even though there are limits on how well they can work, the shown accuracy of recognition and stability of the system suggest a strong foundation for more progress and wider use in both personal and business settings.

The future work on the SignSpeak system will focus on updating the hardware to fix the problems found in the current prototype. The Arduino-based version has shown that the system can work, but it is limited in terms of portability, autonomy, and overall usefulness. To get around these problems, the plan is to switch to a smaller architecture that uses current microcontrollers. This change will make it easier for wireless data to be sent using protocols like Bluetooth or Wi-Fi, which will let the system work on its own without a PC. The focus of the hardware upgrade will be on making an ergonomic glove by using flexible printed circuit boards and lightweight, modern materials. This will make it more comfortable for users and encourage them to keep using it. The data collected from gesture input will be used to improve the training of the machine learning models that are used in the recognition process.

Big technological changes will include using high-capacity lithium-polymer batteries to make batteries last longer, along with energy-efficient parts to make sure they keep working. Adding small, accurate sensors will improve gesture detection accuracy, and a modular design will make it easier to maintain the system and add new features in the future. Along with hardware upgrades, it is suggested that a mobile app be made to serve as the main way for users to interact with the system. The program will have basic features like calibrating devices and translating gestures in real time. In addition, it will have personalized user profiles, usage statistics, and training materials. The app's interface will be designed to be easy to use and understand, even for people with hearing or speech impairments. The program will also have a library of gestures, interactive learning tools, the ability to connect with voice assistants, and cloud storage for user data and settings.

Working with other universities will be very important for making the system bigger and more useful. Working with medical facilities could help the system make it easier for patients to get better and keep an eye on their health. This means making APIs that work with medical record systems and collecting useful data for clinical use.

In schools, the system could help with training programs and improve sign language literacy by working with digital learning platforms and making learning materials. Working with public organizations will make sure that the system really meets the needs of people who have trouble hearing or speaking. These partnerships will help get user feedback and improve the system based on how it is used in real life.

Technological progress will focus on making the software and algorithms of the system better. Future versions will use advanced machine learning algorithms to make it easier to recognize complex or fast movements and to add more sign language vocabulary. The study will investigate new ways of processing signals, deep learning frameworks, and new types of sensors that can improve input fidelity. As adaptive learning processes get better, the system will be better able to handle different signing styles.

At the same time, steps will be taken to make the SignSpeak system more energy-efficient and responsive. The system is wanted to become an affordable, and user-friendly tool for making it easier for people with hearing or speech disabilities to communicate. To do this, it will undergo a full development strategy that includes upgrading hardware, improving software, integrating with medical institutions, and refining algorithms. These improvements will make the system work better in a wider range of situations, such as daily communication, education, and rehabilitation. This will help achieve the main goal of promoting social inclusion and accessibility.

V. CONCLUSION

The SignSpeak system represents a significant step toward inclusive communication technologies tailored for individuals with hearing and speech impairments. Through its low-cost, AI-powered wearable solution, the project demonstrates that effective real-time sign language translation can be achieved without reliance on high-end commercial devices. Results show promising levels of accuracy, performance, and usability, indicating strong potential for practical deployment.

Future developments will focus on enhancing hardware ergonomics, expanding the gesture recognition capabilities, and building a mobile application to facilitate broader adoption. Collaboration with institutions and user communities will ensure the system remains user-centric and impactful. With continued research and integration efforts, SignSpeak holds promise to become a widely accessible assistive technology that bridges communication gaps and promotes social inclusion.

Given the ongoing rise in the number of people with hearing impairments worldwide and limited access to current commercial solutions, the study demonstrated the project's relevance. The prototype system showed acceptable gesture recognition accuracy (over 90% in most scenarios), adequate speed (an average of 112 ms per recognition), and long-term use stability.

The project architecture is built on a distinct separation of hardware and software, including flexible SpectraFlex sensors, an MPU6050 motion sensor, an Arduino microcontroller, as well as software modules written in Python using TensorFlow. The system guarantees adaptability to the needs of a particular user by supporting individual learning, the capacity to customize gestures, and periodic updates of the machine learning model.

The tests showed the system's main benefits: simplicity, scalability, adaptation to new circumstances, and low cost in comparison to analogues. Among the drawbacks are low autonomy, the requirement for user-specific initial calibration, and less accuracy with quick or complicated gestures.

Despite these encouraging findings, the current prototype presents limitations. First, the system is restricted to single-hand gesture recognition, which means that many two-handed signs commonly used in full sign languages cannot yet be translated. Second, while alphabet letters and selected expressions were successfully recognized, the ability to process continuous words or sentences remains untested. These aspects will be considered as directions for future research. Subsequent work will focus on dual-hand integration, expansion of the gesture vocabulary to cover a broader portion of sign language, and refinement of algorithms for continuous gesture recognition. Moreover, portability and autonomy improvements, including wireless communication and energy-efficient hardware, will be pursued to enhance everyday usability.

SignSpeak shows therefore the possibility of developing a creative, accessible, and efficient tool for communication that helps social integration and enhances the quality of life of individuals with hearing or speech disabilities.

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IULIANA MARIN received the B.S. degree in computer science, the M.S. degree in software engineering, and the Ph.D. degree in computers and information technology from the National University of Science and Technology POLITEHNICA Bucharest, Romania, in 2015, 2017, and 2020, respectively. She is currently an Associate Professor and a Ph.D. Coordinator with the University of Science and Technology POLITEHNICA Bucharest and she has expertise in databases, operating systems, computer networks, the semantic web, distributed systems, technologies for big data analysis, and the Internet of Things. She received several prizes, including the Gold Medal for the invention "Preeclampsia and Detection of Secondary Diseases Based on Heart Rate and a Weighted Stratified Network" at the Salon of Scientific Research, Innovation and Inventions "PRO INVENT," Cluj-Napoca, Romania, in 2020, and the Silver Medal for the invention "Automatic System for Establishing the Diet in Case of Preeclampsia" at the International Exhibition of Inventions and Innovations "Traian Vuia," Timişoara, Romania, in 2019.

VOLODYMYR BABALIAN received the M.D. degree from Kharkiv National Medical University (KHNMU), Kharkiv, Ukraine, the Ph.D. degree in traumatology and orthopedics, in 2006, and the Doctor of Science degree in traumatology and orthopedics (medicine), in 2022. He is currently a Professor with the Department of Traumatology, Orthopedics, Physical and Medical Rehabilitation, KHNMU, and a Practicing Traumatologist Surgeon at the Orthopedic and Traumatology Department of the University Clinic of KHNMU. He has expertise in joint endoprosthetics, arthroscopy, osteosynthesis, reconstruction of bone deformities, and the prevention of injuries and joint diseases in both civil and military fields. He has 25 years of clinical experience and is the co-author of 20 patented inventions and over 100 scientific articles. He is a member of American Association of Orthopedic Surgeons (AAOS).

SERHII SLUTU is currently pursuing the bachelor's degree in computer science from the Faculty of Engineering in Foreign Languages, National University of Science and Technology POLITEHNICA Bucharest, Romania, supported by a grant awarded by the Ministry of Education of Romania. Since 2021, he has been enrolled in the Bachelor's Program in Computer Engineering at Kharkiv National University of Radio Electronics, Ukraine. He has experience in both software and data engineering, with a focus on innovative applications of artificial intelligence (AI). His research interests include AI and bioengineering. He is a member of European Society of Human Genetics (ESHG).